

$$\text{Given, } \vec{a} \times (\vec{b} \times \vec{a}) = \alpha \hat{i} - \beta \hat{j} + \gamma \hat{k}$$

Compare that with (i), we get

$$\alpha = 5, \beta = -4, \gamma = -1$$

$$\therefore \alpha + \beta + \gamma = 5 - 4 - 1 = 0$$

67. **Option (a) is correct.**

Let a vector $\vec{a}$ have (a_1, a_2, a_3) components

$$\Rightarrow a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$$

$$\therefore |\vec{a}| = 2$$

Also, it is given that $\vec{a}$ make angle with $\frac{\pi}{3}$ with

$$2\hat{i}, \frac{\pi}{4} \text{ with } 3\hat{j} \text{ and an acute angle } \theta \text{ with } 4\hat{k}$$

Then, we have

$$\vec{a} \cdot 2\hat{i} = |\vec{a}| |2\hat{i}| \cos \frac{\pi}{3}$$

$$(a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}) \cdot 2\hat{i} = 2 \times 2 \times \frac{1}{2}$$

$$2a_1 = 2$$

$$a_1 = 1$$

$$\therefore \vec{a} \cdot 3\hat{j} = |\vec{a}| |3\hat{j}| \cos \frac{\pi}{4}$$

$$3a_2 = 2 \times 3 \times \frac{1}{\sqrt{2}}$$

$$a_2 = \sqrt{2}$$

$$\text{and } \vec{a} \cdot 4\hat{k} = |\vec{a}| |4\hat{k}| \cos \theta$$

$$4a_3 = 2 \times 4 \cos \theta$$

$$\cos \theta = \frac{a_3}{2}$$

$$\text{Now, } |\vec{a}| = 2$$

$$\sqrt{a_1^2 + a_2^2 + a_3^2} = 2$$

Square both side, we get

$$a_1^2 + a_2^2 + a_3^2 = 4$$

Put the value of $a_1 = 1$ and $a_2 = \sqrt{2}$

$$(1)^2 + (\sqrt{2})^2 + a_3^2 = 4$$

$$\Rightarrow a_3^2 = 4 - 1 - 2$$

$$\Rightarrow a_3^2 = 1$$

$$\Rightarrow a_3 = \pm 1$$

$$\text{When } a_3 = -1 \text{ then } \cos \theta = \frac{-1}{2}$$

$$\Rightarrow \theta = \frac{2\pi}{3} > \frac{\pi}{2} \quad (\text{contradict})$$

$\therefore \theta$ is acute angle with $4\hat{k}$

$$\text{So } a_3 = 1$$

Hence, the components of $\vec{a} = (1, \sqrt{2}, 1)$

68. **Option (b) is correct.**

Statement 1 is wrong.

$$\text{We know that } \vec{\tau} = \vec{r} \times \vec{F}$$

The moment of force about a point is dependent of application of force

Statement 2 is correct.

The moment of force about a line is a vector quantity.

Cross product of two vector is again a vector

$$\begin{array}{ccc} \vec{\tau} & = & \vec{r} \times \vec{F} \\ \downarrow & \downarrow & \downarrow \\ \text{vec} & \text{vec} & \text{vec} \end{array}$$

69. **Option (a) is correct.**

Given

$$(\vec{r} \cdot \hat{i})(\vec{r} \times \hat{i}) + (\vec{r} \cdot \hat{j})(\vec{r} \times \hat{j}) + (\vec{r} \cdot \hat{k})(\vec{r} \times \hat{k}) \quad \dots(i)$$

We know that

$$\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$(\vec{r} \cdot \hat{i}) = (a\hat{i} + b\hat{j} + c\hat{k}) \cdot \hat{i} = a$$

$$(\vec{r} \cdot \hat{j}) = b$$

$$(\vec{r} \cdot \hat{k}) = c$$

$$\begin{aligned} \text{and } \vec{r} \times \hat{i} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ 1 & 0 & 0 \end{vmatrix} \\ &= -\hat{j}(-c) - b\hat{k} \\ &= c\hat{j} - b\hat{k} \end{aligned}$$

$$\text{Similarly } \vec{r} \times \hat{j} = -c\hat{i} + a\hat{k} \text{ and}$$

$$\vec{r} \times \hat{k} = b\hat{i} - a\hat{j}$$

Now put all these values in equation (i), we get

$$\begin{aligned} &a(c\hat{j} - b\hat{k}) + b(-c\hat{i} + a\hat{k}) + c(b\hat{i} - a\hat{j}) \\ &= ac\hat{j} - ab\hat{k} - cb\hat{i} + ab\hat{k} + bc\hat{i} - ac\hat{j} \\ &= \vec{0} \end{aligned}$$

70. **Option (b) is correct.**

The angle between $\vec{a}$ and $\vec{a} - \vec{b}$ is given by

$$\cos \theta = \frac{\vec{a} \cdot (\vec{a} - \vec{b})}{|\vec{a}| |\vec{a} - \vec{b}|}$$

$$\text{Now, } \vec{a} \cdot (\vec{a} - \vec{b}) = \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b}$$